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A CT-based deep learning model to differentiate between benign and malignant adrenal lesions

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ABSTRACT

Objective: To develop a deep learning model to differentiate benign from malignant adrenal lesions.

Materials and methods: A total of 380 patients with 385 pathologically confirmed adrenal lesions (101 malignant, 284 benign) were retrospectively included. Adrenal lesions were manually segmented on CT images and analyzed in a deep learning pipeline aimed at differentiating benign from malignant lesions. Four predictive models were developed that incorporated combinations of radiological data (tumor size, and spontaneous attenuation) and non-radiological data (*i.e.*, medical history and laboratory results). Data of 267 patients were used as a training set and those of 113 patients for the test set. The diagnostic capabilities of the four models were estimated using sensitivity, specificity, accuracy, and areas under the receiver operating characteristic curves (AUC) using histopathological findings as the gold standard. The reproducibility of manual segmentation was estimated using the Dice similarity coefficient after blinded resegmentation of 40 adrenal lesions by an independent radiologist.

Results: Segmentation reproducibility achieved a mean Dice similarity coefficient of 0.92 ± 0.03 (range: 0.72–0.97). The most accurate model, which combined clinical, biological, and radiological data, achieved 84.2% accuracy (95% confidence interval: 79.9, 88.6) and an AUC of 0.93 (95% confidence interval: 89.9, 97.9) in the test set for diagnosis of malignant adrenal lesion.

Conclusion: A deep learning model integrating preoperative clinical, biological, and radiological features demonstrates high capabilities in differentiating benign from malignant adrenal lesions on initial CT examination.

1. Introduction

Adrenal lesions primarily consist of incidentalomas, which are found in 3 to 7% of routine abdominal imaging examinations [1]. With the exception of adrenal lesions that display typical imaging features, distinguishing benign from malignant lesions remains challenging in many patients [1–3]. In this regard, it is estimated that 20% of adrenalectomies performed due to suspected malignancy could be avoided [4,5]. Consequently, the management of patients with adrenal lesions often requires multiple imaging and biological tests.

To improve the preoperative, non-invasive evaluation of patients with adrenocortical carcinoma (ACC), the 2023 guidelines from the European Network for the Study of Adrenal Tumors (ENSAT) suggest that radiomics and artificial intelligence tools may be helpful [1].

Abbreviations: AA, Adrenal adenoma; ACC, Adrenocortical carcinoma; AUC, Area under the receiver operating characteristic curve; CI, Confidence interval; CT, Computed tomography; DL, Deep-learning; DSC, Dice similarity coefficient; DFS, Disease-free survival; ENSAT, European network for the study of adrenal tumors; HU, Hounsfield unit; LPAA, Lipid-poor adrenal adenoma; OS, Overall survival; SD, Standard deviation.

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Radiomics consists in the analysis of mathematically derived textural features on radiologic imaging [6] and has demonstrated capabilities in the field of adrenal lesion characterization [6–8]. Radiomics can also be used to predict overall survival (OS) and disease-free survival (DFS) in other cancers [9,10]. In addition, the use of deep learning (DL) may improve the performance of artificial intelligence-based approach to characterize adrenal lesions.

DL uses several layers of artificial neurons to mimic human perception in recognizing features from input data [11]. In a recent study, DL showed high accuracy for the diagnosis of adrenal adenoma among indeterminate adrenal lesions and yielded 90% sensitivity (95% confidence interval [CI]: 82, 95), 100% specificity (95% CI: 100, 100), and an area under the receiver operating characteristic curve (AUC) of 0.99 (95% CI: 0.97, 1.00) [12]. Other studies reported high accuracy of DL for discriminating between ACC and large adrenal adenomas [13] or for differentiating between adrenal metastases and benign adrenal lesions [14]. Recently, a dominant trend in the characterization of adrenal lesions has emerged, which consists in combining imaging features and clinical-biological characteristics in order to improve characterization [15]. However, to date, no study has specifically evaluated a DL algorithm incorporating clinical, biological, and imaging features for the differentiation between benign and malignant adrenal lesions.

The purpose of this study was to evaluate a DL algorithm combing computed tomography (CT) images and clinico-biological data for distinguishing between benign and malignant adrenal lesions.

2. Materials and methods

2.1. Patients

This study was approved by our internal review board. All included patients gave informed consent. The database of our institution was quiered to identify all patients who underwent adrenalectomy between 2007 and 2023 with a histopathologically confirmed diagnosis of ACC, adrenal metastases, adrenal adenoma (AA), or pheochromocytoma. Patients were further eligible when they underwent an abdominal CT examination less than three months prior to surgery, that included unenhanced images and images obtained during the venous phase following intravenous administration of iodinated contrast material. Patients were excluded when no clinico-biological data were available or when CT examinations were incomplete (Fig. 1).

2.2. Clinical and biological data

Baseline patient characteristics were retrospectively extracted from medical records. They included demographic data (age, sex, height, weight, body mass index, presence of diabetes, and presence of hypertension, with hypertension defined as a systolic blood pressure above 140 mmHg or a diastolic pressure above 90 mmHg), tumor characteristics (lateralization and histological subtype) and biological features (presence of hormonal secretion, and type of hormone).

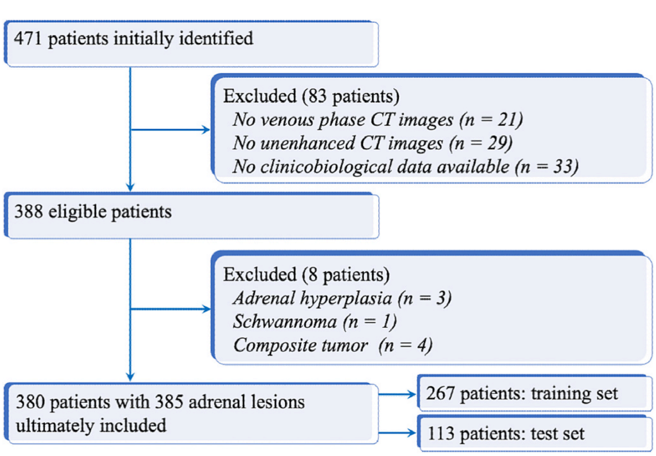

Fig. 1. Study flow chart.

2.3. Definition of the ground truth

The results of histopathological examination performed after adrenalectomy were used as the standard of reference to classify adrenal lesions. Adrenal metastases, ACCs, and pheochromocytomas with distant metastases at the time of diagnosis or that developed during the follow-up were classified as malignant. AA and pheochromocytomas without distant metastases at diagnosis or during the follow-up were classified as benign.

2.4. CT image acquisition

CT images were acquired from with three different CT scanners including Revolution HD® (General-Electric Healthcare, Wauwatosa, WI, USA), Somatom Sensation® 64 (Siemens Healthineers, Erlangen, Germany), or Somatom Definition® Flash (Siemens Healthineers). All CT examinations included images obtained before and after intravenous administration of an iodinated contrast material containing 300 to 350 mg of iodine per mL, at a rate of 1.5–2 mL/kg, during the venous phase of enhancement (65–75 s after the start of contrast material administration) and were reconstructed with a standard kernel using a 512 × 512 matrix and a slice thickness ranging from 0.6 to 3 mm.

2.5. Image analysis

One radiologist (M.B., with 10 years of experience in abdominal imaging), blinded to the clinical outcomes reviewed all CT examinations after anonymization on a picture-archiving and communication system viewing station (DirectView®, 11.4.0.1253 sp1 version, Carestream Health, Rochester, NY, USA). For all adrenal lesions, the radiologist measured the largest length, width, and height, and the mean spontaneous attenuation in Hounsfield units (HU) by placing a region of interest on the largest portion of the lesion (at least two-thirds of the lesion).

Then, all CT images were converted to Neuroimaging Informatics Technology Initiative format. Adrenal lesions were manually segmented on each converted CT image by the same radiologist using a dedicated tool (3DSlicer software, available at https://www.slicer.org) [16]. To assess the reproducibility of the segmentation process, a second radiologist (ZH, with five years of experience in abdominal imaging) independently segmented the same adrenal lesions on 40 randomly selected cases.

2.6. Development of the deep-learning models

The study cohort was randomly divided into a training set and a test set using an algorithm. The training set included 267 patients (68 with malignant lesions and 199 with benign lesions), and the test set included 113 patients (31 with malignant lesions and 82 with benign lesions). For all adrenal lesions, preprocessing was made by extracting the portion of the CT images that contained the entire adrenal lesion and applying a 64 × 128 × 128 pixels patch. The reconstruction slice thickness was set to a 2 × 2 × 2 mm^3 matrix. The patch was centered on the adrenal lesion using segmentation on all CT scans. Voxel value normalization was performed within a range of attenuation values between −350 HU and 450 HU.

Four DL models were built. Models 1 and 2 included human interpretation and models 3 and 4 were fully automated and did not require human intervention. Model 1 was developed using only features collected by an expert radiologist (M.B., with fifteen years of experience in adrenal imaging) from a tertiary center. These features consisted of the largest adrenal lesion diameter measured in three planes (length, width, and height), and the mean spontaneous attenuation value in HU [1]. Classification was performed using a three-layer multilayer perceptron with four input features, 1024 features in the hidden layers and two output features for malignant or benign classification (Fig. 2). A LeakyRELU activation function with a negative slope of 0.01 was used at the output of the input layer and at the output of the first hidden layer (Fig. 2). Model 2 was developed similarly as Model 1 with the addition of clinico-biological features. Model 3 was developed using CT data and segmentation masks of the adrenal lesion. The classification was performed using a two-channel, three-dimensional convolutional encoder inspired by the U-Net architecture [17]. This encoder was composed of six steps, each with two convolutions. The three-dimensional convolution was made from kernels of size 3 × 3 × 3 mm^3. Due to the large amount of data, a batch size of 8 was chosen. With this batch size, instance normalization was preferred over batch normalization because instance normalization is independent of the batch size. Skip connections were implemented to counteract vanishing gradients. Dropouts were not chosen. Then, a single-layer perceptron with 320 input features and two or four output classes was used. Model 4 was developed similarly to Model 3, with the addition of clinico-biological features (Fig. 2).

Each model was trained to differentiate between malignant and benign adrenal lesions. The loss function employed was a weighted cross-entropy loss with a weighting ratio of 1:3, reflecting the proportion of malignant to benign adrenal lesions in our cohort. The maximum number of training epochs was set to 10,000, and early stopping was implemented if the learning rate decreased below 0.0001. To reduce the risk of overfitting and improve model generalization capabilities, data augmentation methods were added, including noise addition and image rotations.

2.7. Statistical analysis

All statistical analyses were conducted using Python (version 3.12.3). Qualitative variables were expressed as raw numbers, proportions and percentages. Quantitative variables were expressed as means, standard deviations (SD), and ranges for normally distributed variables, or as medians, first and third quartiles and ranges, for those that did not follow a normal distribution [18]. Comparisons were made using Wilcoxon rank-sum test or Student's *t*-test for quantitative data, and using Chi-square test for qualitative data.

Assessment of the interobserver reproducibility of the manual segmentations of the tumors was performed using the Dice similarity coefficient (DSC) [19]. DSC between 0.00 and 0.20; 0.21 and 0.40; 0.41 and 0.60; 0.61 and 0.80; and 0.81 and 1.00 indicated slight, fair, moderate, substantial, and almost perfect agreement, respectively [9].

Evaluation of the diagnostic performances of the DL models was performed with several metric. For all models, diagnostic performance for the differentiation between benign and malignant adrenal lesions was assessed using sensitivity, specificity, and accuracy. Receiver operating characteristic (ROC) curves were generated for each model, and the area under the ROC curve (AUC) values were calculated. Comparisons of AUCs between models were performed using the DeLong method. Differences in sensitivity, specificity and accuracy between models were searched for using the McNemar test. A p-value $<$ 0.05 was considered to indicate statistically significant differences.

3. Results

3.1. Patients

Of the 471 patients who were initially considered for eligibility, 380 were ultimately included in the study. Fig. 1 shows the study flowchart. There were 129 men and 251 women with a mean age of 53.0 ± 14.8 (SD) years (age range: 18.7–91.6 years) with a total of 385 adrenal lesions (five patients had bilateral adrenal lesions). Of the included lesions, 101 (26.2%) were malignant adrenal lesions and 284 (73.8%) were benign adrenal lesions.

Malignant adrenal lesions consisted of 87 ACCs, 11 adrenal metastases and three pheochromocytomas. Benign adrenal lesions consisted of

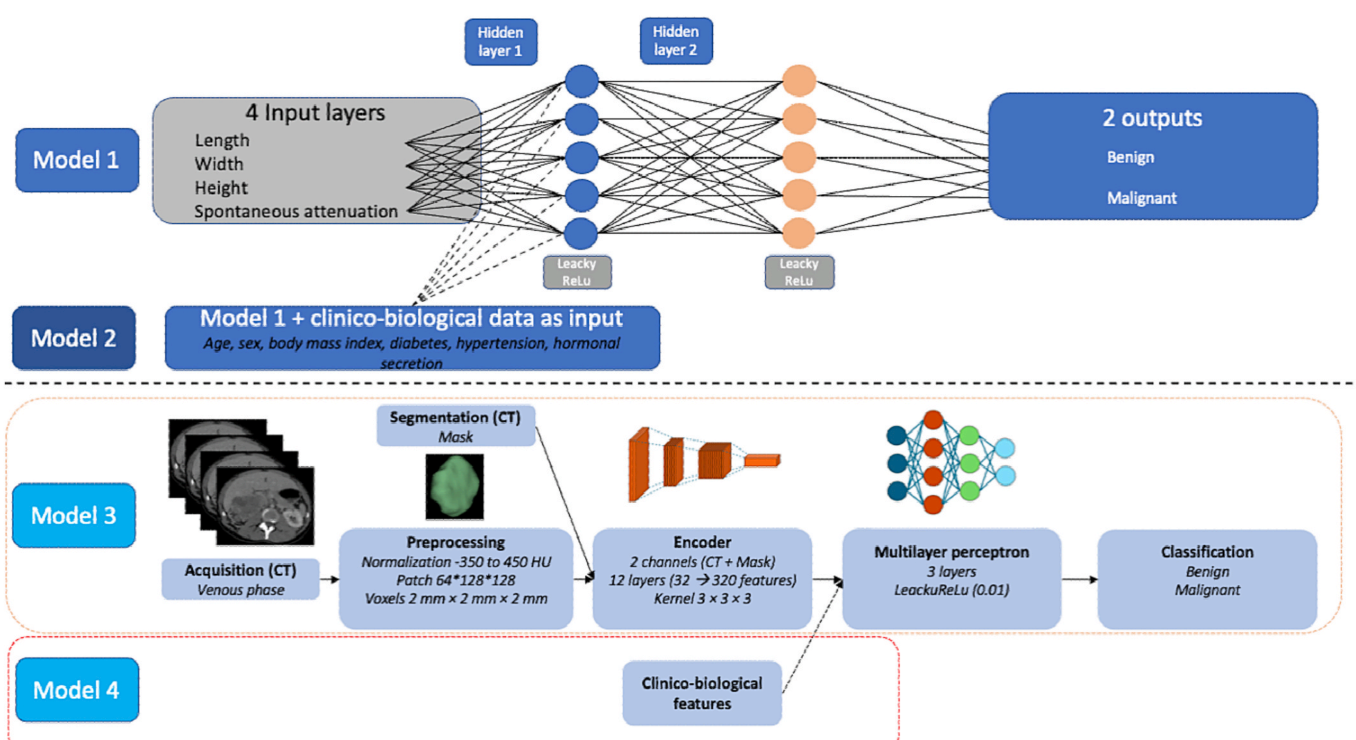

Fig. 2. Diagram shows the four models used in the study to differentiate between malignant and benign adrenal lesions. CT indicates computed tomography. HU indicates Hounsfield unit.

165 AA (including 22 lipid rich AA with a spontaneous attenuation < 10 HU and 143 lipid-poor AA with a spontaneous attenuation > 10 HU on unenhanced CT images) and 119 pheochromocytomas. The definition of benign pheochromocytoma was established with a mean follow up period of 51 ± 46 (SD) months (range: 15–206 months) after surgery.

Considering the clinico-biological features, no significant differences were found in the prevalence of diabetes (p = 0.339) or high blood pressure (p = 0.151) between patients with benign adrenal lesions and those with malignant adrenal lesions. The initial characteristics of patients and adrenal lesions including age, sex, body mass index, laterality, maximal diameter, and hormonal secretion status are reported in Table 1.

3.2. Adrenal lesion segmentation reproducibility

Segmentation was performed for all adrenal lesions. Considering the 40 patients randomly selected and the resegmented adrenal lesions, the mean DSC was 0.92 ± 0.03 (95% CI: 0.72, 0.97). It was considered almost perfect (≥ 0.81) for 39/40 (98%) adrenal lesions and substantial (0.61 to 0.80) for one adrenal lesion (2%).

3.3. Adrenal lesion characterization

Model 1, which included the largest diameter measured in three planes (length, width, and height) and the mean spontaneous attenuation yielded 37% sensitivity (95% CI: 25.4, 47.8), 89.2% specificity (95% CI: 85.0; 93.6), 75.2% accuracy (95% CI: 70.1, 80.5), and an AUC of 0.752 (95% CI: 0.681, 0.823) for the diagnosis of malignant adrenal lesions in the test cohort.

Model 2 had the highest performances, with 81.5% sensitivity (95% CI: 72.7, 0.90.7), 93.2% specificity (95% CI: 89.9, 96.9), 90.3% accuracy (95% CI: 86.7, 93.8), and an AUC of 0.951 (95% CI: 0.915, 0.987) for the diagnosis of malignant adrenal lesion in the test cohort.

Model 4, which included DL classification from a mask of segmentation and clinico-biological features without any human participation had 40.7% sensitivity (95% CI: 29.4, 52.2%), 100% specificity (95% CI: 98.1, 100), 84.2% accuracy (95% CI: 79.8, 88.6), and an AUC of 0.939 (95% CI: 0.900, 0.980) for the diagnosis of malignant adrenal lesion in the test cohort Table 2.. No significant difference in AUC was found between Model 2 and Model 4 (p = 0.490) (Table 3., Fig. 3).

4. Discussion

In this study, we developed and compared four DL algorithms to differentiate between benign and malignant adrenal lesions. Two key findings emerged. First, incorporating clinical and biological data into radiological data, as in Model 2 and Model 4, significantly improved diagnostic performance compared to analyzing CT data alone. Second, we found that a fully automated DL model achieved diagnostic performance similar to that of a model that used CT features extracted by an expert radiologist from a tertiary center specializing in adrenal lesions.

Table 1
Patients and adrenal lesions characteristics.

Variables	Malignant AL (n = 101)	Benign AL (n = 284)	p-value*
Age (year)	54.8 (41.6, 66.3)	52.6 (42.8, 61.7)	0.351
Female	70 (69.3%)	183 (64.4%)	0.458
Body mass index	26.0 (22.6, 29.7)	26.0 (22.6, 30.2)	0.867
Adrenal secretion	64 (63.4%)	233 (82.0%)	**0.014**
Cortisol	50 (49.5%)	9 (31.3%)	**0.002**
Aldosterone	5 (5.0%)	32 (11.3%)	0.119
Catecholamines	6 (5.9%)	113 (39.8%)	**0.001**
Androgenes	32 (31.7%)	3 (1.1%)	**0.001**
Larger diameter (mm)	90.0 (60.0, 140.0)	40.0 (26.3, 55.0)	**0.001**
Side (Left/Right)	55 (54.5%)/46(45.5%)	148(52.1)/136 (47.9)	0.685

Quantitative variables are expressed as medians followed by first and third quartiles into parentheses. Qualitative variables are expressed as proportions followed by percentages into parentheses.*Numbers in bold indicats p-value < 0.05.

Table 2
Diagnostic performances of the four models in differentiating between benign and malignant adrenal lesions.

	Sensitivity (95% CI) (%)	Specificity (95% CI) (%)	Accuracy (95% CI) (%)	AUC (95% CI)
Model 1	37 (25.4–47.8)	89.3 (85.0%–93.6)	75.3 (70.1–80.5)	74.8 (67.6–82.0)
Model 2	81.6 (72.7–90.7)	93.4 (89.9–96.9)	90.2 (86.7–93.8%)	95.1 (91.5–98.7)
Model 3	44.4 (36.0–52.8)	82.7 (77.4–88.0)	72.3% (66.9–77.7%)	71.4 (64.4–78.4)
Model 4	40.7 (29.4–52.3)	100.0 (98.1–100.0)	84,2% (79.9–88.6)	93.9% (89.9 –97.9)

Table 3
Results of the comparisons of areas under the ROC curves of four different models to characterize adrenal lesions.

	Model 1	**Model 2**	**Model 3**	**Model 4**
Model 1	1	–	–	–
Model 2	**0.002**	1	–	–
Model 3	0.361	**0.002**	1	–
Model 4	**0.003**	0.490	**0.003**	1

Numbers represent the p-values obtained using the DeLong test to compare the areas under the ROC curves of all the models.

Previous studies have shown that the sensitivity for the diagnosis of benign adrenal lesions using imaging features and radiologist analysis alone (*i.e.*, without AI) reaches 70% [10,20]. However, most of these studies included only two different histological subtypes of adrenal lesion, mostly metastases and benign lipid rich AA [10,20]. By contrast, our study included four different adrenal histological subtypes. Singh et al. achieved 91% accuracy using AI to distinguish between benign and malignant adrenocortical lesions and to differentiate stage 1–2 ACCs from lipid-poor adrenocortical adenomas (LPAAs). However, their studyincluded only 48 ACCs and 43 LPAAs [13]. Kusunoki et al. aimed to diagnose AA alongside other histological subtypes. Using a deep convolutional neural network model, these researchers achieved sensitivity and specificity > 90%, as well as an AUC > 0.90, [12]. Similarly, Wang et al. built six DL models that could differentiate LPAAs from adrenal metastases with an AUC > 0.8 [21]. However, like other published studies that aimed to differentiate adrenal metastases from AAs [14], or pheochromocytomas and AAs [22,23], these algorithms have high performances but are not fully clinically relevant. In this regard, most adrenal metastases appear during the follow-up of patients with known cancer, and most pheochromocytomas secrete metanephrines. This often makes the diagnosis straightforward, eliminating the need for AI [24].

Despite the ENSAT guidelines recommending against biopsy and surgery for most benign adrenal lesions, no study to date has included non-histologically proven adrenal lesions. [1]. Including histologically unproven adrenal lesions could dramatically increase cohort sizes. However, incorporating uncertain features into AI models introduces significant methodological flaws, so this approach should not be considered [16].

Recently, other applications of AI in adrenal imaging have emerged. For example, Robinson-Weiss et al. developed a machine-learning algorithm that can automatically segment adrenal glands on contrast-enhanced CT images and classify them as normal or mass-containing [25]. Their model achieved 69% sensitivity (95% CI: 58, 79) and 91% specificity of (95% CI: 90, 92) for the diagnostic of AL [25]. Xi et al.

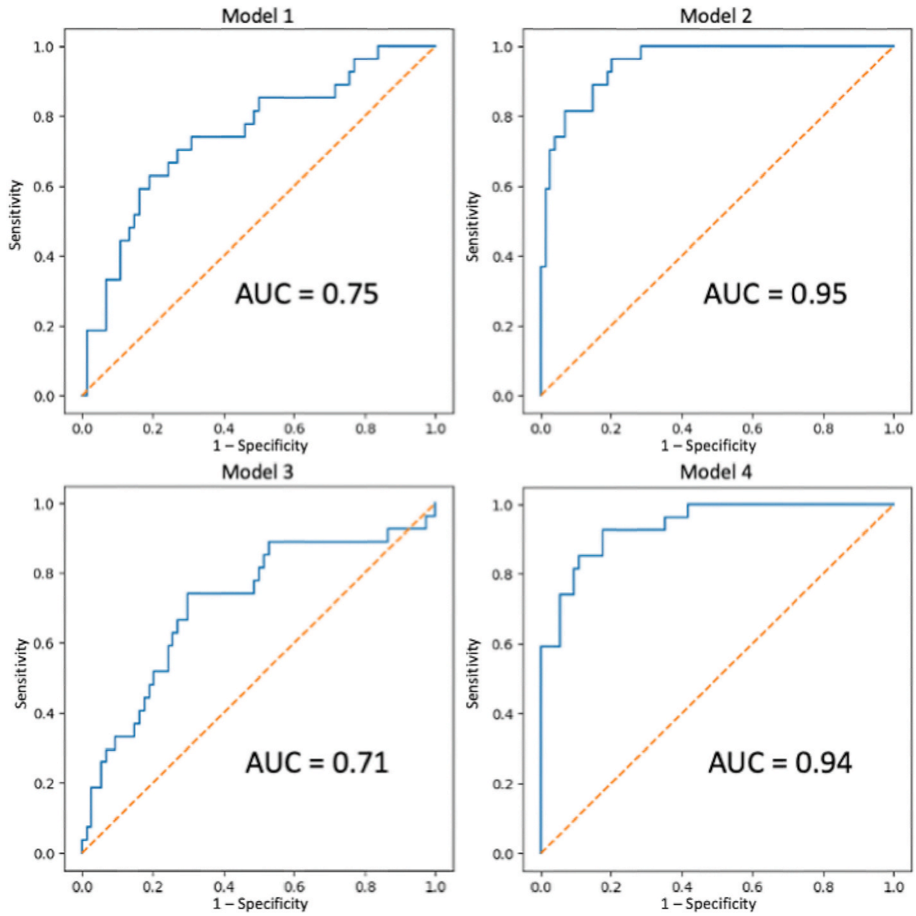

Fig. 3. Graphs show the areas under the receiver operating characteristic curves (AUC) of the four models to differentiate between benign and malignant adrenal lesions. Models 1, 2, 3 and 4 shows AUCs 0.75, 0.95, 0.71 and 0.94, respectively of.

obtained similar using a training cohort of 443 patients and a multi-center test cohort of 335 patients [26]. They obtained an AUC of 0.88 for AL detection using a cascade network based on a deep learning algorithm [26]. These results are an important step toward developing a complete AI algorithm that would include detection, segmentation, analysis, and classification [24]. Bai et al. proposed a multiclass segmentation model that can differentiate LPAA from nodular hyperplasia with 91.7% accuracy (95% CI: 76.8–98.9%) on an external test set [27]. Another approach was proposed by Romeo et al., who developed a machine learning model that uses texture analysis features from magnetic resonance images to distinguish between lipid-rich adrenal adenomas, lipid-poor adrenal adenomas, and non-adenoma adrenal lesions [28]. The model yielded 80% accuracy, similar to that of an expert radiologist (73%) ($p = 0.13$) [28]. Future studies could explore developing an algorithm to classify different histological subtypes of adrenal lesions.

Our study has some limitations. First, the retrospective, single center design requires cautious interpretation. However, due to the rarity of primitive adrenal tumors such as ACC, which have an incidence rate of one to two cases per million people per year, it would take many years to recruit enough patients for a prospective study to train an AI model. Second, we only included histologically proven adrenal lesions. As previously described, a diagnosis must be made before data can be entered into an AI model, because training a model with inaccurate data could compromise its interpretability. Furthermore, we did not consider an external test cohort, as recommended by van Griethuysen et al. [16]. Finally, we considered only four histopathological types: AA, ACC, pheochromocytoma, and metastases, omitting other rare subtypes. However, these four histological subtypes account for over 95% of adrenal lesions and restricting the number of histological types may improve the stability of the model and extrapolation for daily use [1].

In conclusion, our fully automated DL model, which integrates preoperative clinical, biological, and radiological features demonstrates high capability in differentiating benign from malignant adrenal lesions on initial CT examination in a test cohort. While future prospective validation is needed, this study confirms that AI can help physicians improve the early management of patients with adrenal lesions.

CRediT authorship contribution statement

Zack Huang: Writing – original draft, Investigation, Formal analysis, Data curation, Conceptualization. **Anthony Dohan:** Writing – review & editing, Methodology, Conceptualization. **Guillaume Assié:** Writing – review & editing, Validation, Methodology, Conceptualization. **Martin Gaillard:** Supervision, Data curation. **Florian Violon:** Validation, Data curation. **Anne Jouinot:** Supervision, Methodology, Data curation. **Philippe Soyer:** Writing – review & editing, Methodology, Conceptualization. **Jérôme Bertherat:** Visualization, Supervision, Conceptualization. **Rafael Marini:** Software, Resources, Methodology. **Guillaume Chassagnon:** Writing – original draft, Methodology, Conceptualization. **Maxime Barat:** Writing – review & editing, Supervision, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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